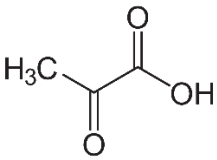
Analytical Chemistry Cumulative Exam
March 10, 2023

Some useful equations and other data.

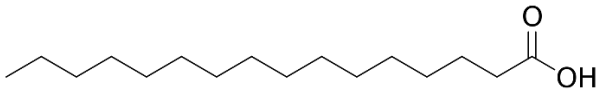
$$u_r = \sqrt{{u_{x1}}^2 + {u_{x_2}}^2 + \ldots} \qquad \frac{u_r}{R} = \sqrt{\left(\frac{u_{x1}}{R_{x1}}\right)^2 + \left(\frac{u_{x2}}{R_{x2}}\right)^2 + \ldots} \qquad \mu = \overline{x} \pm \frac{ts}{\sqrt{n}} \qquad \mu = \frac{1}{2}\sum c_i {z_i}^2$$

$$-\log \gamma_i = 0.5{z_i}^2\sqrt{\mu} \qquad [H^+] = \sqrt{\frac{K_1K_2F + K_1K_w}{K_1 + F}} \qquad pH = \frac{pK_{a1} + pK_{a2}}{2}$$

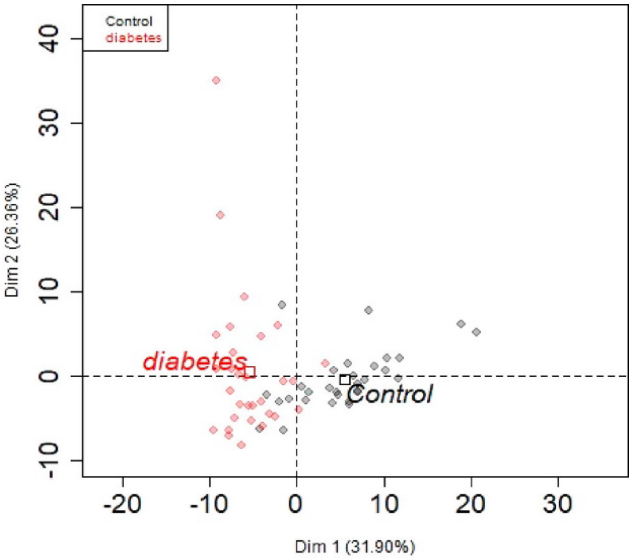
$$E = E^0 - \frac{0.059}{n}\log(Ratio)$$

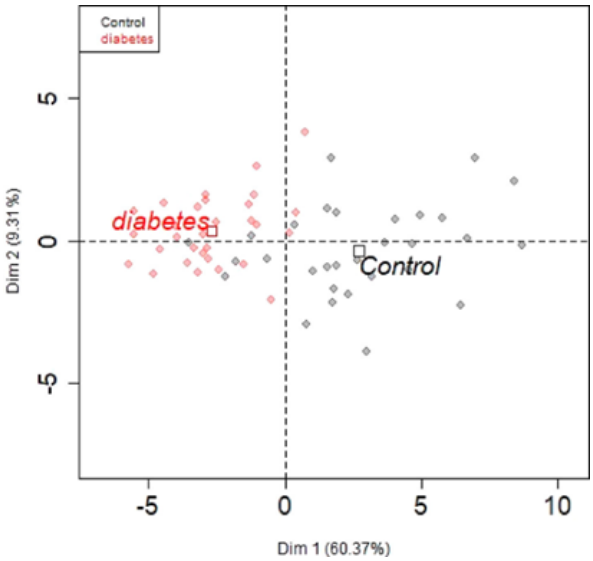


Pyruvic Acid



Palmitic Acid





LEFT: Score plot using concentrations of 163 lipids for 30 healthy patients (control) and 30 diseased patients (diabetes). RIGHT: Score plot using concentrations of 22 lipids for 30 healthy patients (control) and 30 diseased patients (diabetes).

Data from: DOI: (10.1021/acs.jchemed.9b00924)

March 10, 2023 || Analytical Chemistry Cumulative Exam || Basic Problem Solving/General Awareness of the Literature.

Pick 5 questions to solve from the following 8.

1. Show how you would make a buffer of acetic acid, pH 5, that has an ionic strength (μ) of 0.2 M. For convenience, assume the volume needed is 1.0 L. The pK_a for acetic acid is 4.75. Use molar concentrations in your answer.

2. What is the pH of 0.1 M Na_2HPO_4? For phosphoric acid, the pK_a values are 2.15, 7.2, and 12.4

3. A friend who teaches environmental engineering laboratories at another university recently told me her students became nearly paralyzed when they had to use a 2.000-mL rather than a 3.000-mL volumetric pipet to quantitatively transfer 6 mL. Assume the error tolerance is ±0.006 mL for both pipets. Calculate the error for the use of both pipet scenarios for this procedure. Comment on the change in relative error between a 2 vs 3-mL pipet for this work.

4. What is an aptamer? Why have they been of interest in analytical chemistry for at least 30 years? What other technology might they disrupt?

5. Analytical chemistry graduate students should be reading the journal, *Analytical Chemistry*, for papers related to their research as well as the reviews/perspective articles. In a recent article, the group of Ian Wilson discusses issues with identifying metabolomic peaks. In these characterizations, both liquid chromatographic information as well as mass spectral data are obtained. An issue that arises is that molecules assigned based upon mass spectral data then do not make sense with respect to their retention times observed in the liquid chromatography (reversed phase) system. Two retention time examples given were pyruvic acid (6.69 min) and palmitic acid (6.83 min) and the chemical structures are given in your data handout. Why would these retention times be suspect? What other information could you use that is easily found in databases such as the RSC's ChemSpider to justify your concern that the retention data doesn't make sense?

6. Frequently in bioelectrochemistry, formal standard reduction potentials are used. Technically formal potential can be at any specified pH, but often pH 7 is used. Consider the biological redox cofactor NAD+/NADH (this is not an acidic molecule), calculate the formal potential at pH 7. The standard reduction potential is $NAD^+ + H^+ + 2e^- \rightarrow NADH \quad E_o = -0.320\ V$

7. A recent organic chemistry faculty candidate, Dr. Shang Jia, discussed making new fluorescent dyes for measurements in biological systems. There are many dyes with high quantum yield that fluoresce in the 500 to 600 nm range. Why can't these dyes be used in a biological system? What is the optical wavelength range that is typically needed for biological imaging? Also, define quantum yield and explain why this is an important parameter in biological fluorescence imaging.

8. Principal Components Analysis (PCA) is commonly used in analytical chemistry. You get both a scores and loadings plot from a PCA. However, usually only a scores plot is shown in most manuscripts. In your data/equations table, two plots are shown. In one, there were 163 lipids quantified. In the second, a variety of statistical tests were performed and the data set was reduced from 163 to 22 lipids.

    a. Briefly explain what PCA is and why it is so commonly used in analytical chemistry.
    b. What does Dim 1 and Dim 2 mean? Which one is better the 163 lipids or the 22 lipids? Why?
    c. Why do you think the statistical tests were performed or what do you think the authors were looking to accomplish when reducing the data set from 163 unique lipids to 22 lipids?